Support Vector Machine

Outline

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Managerial motivation

Introduction: Managerial motivation

- The Support Vector Machine connects convex optimization directly to machine learning
- The classifier is obtained by solving a **QP**, with a clear geometric reading
- Immediate managerial applications: credit risk, churn, predictive maintenance, quality control, fraud

No machine learning libraries

- In this chapter we do *not* use ML libraries: every model is a QP written and solved with Gurobi
- The goal is to understand *what* a classifier optimizes
- **A modelling note**: the labels $y_i \in \{-1, +1\}$ are *observed data*, not variables
- The variables (coefficients, intercept, slacks, multipliers) are all continuous

The chapter closes the circle

- The same tools used for production, logistics and finance — convex QPs, duality, multipliers — are the engine of one of the most elegant classifiers of machine learning
- Understanding the SVM as an optimization problem, and not as a black box from a library, gives full control over what the model is really doing

In this chapter we will learn to

1. formulate hard and soft margin as QPs
2. derive and solve the dual, identifying the support vectors
3. obtain non-linear boundaries with kernels while staying in the convex world
4. handle unbalanced classes with asymmetric costs and evaluate with the right metrics

Summary

- The SVM is a classifier obtained by solving a convex QP
- The labels y_i are data; all the variables are continuous
- Every model of the chapter is written and solved with Gurobi, without ML libraries

Hard margin: the geometry

Introduction: Hard margin

- We start from the simplest case: two classes separable by a hyperplane
- Among all separating hyperplanes we look for the one with the widest **margin**
- We will see that maximizing the margin is a convex QP with linear constraints

The problem in words

- *We decide* the coefficients of a hyperplane that separates two classes of observations
- *The objective* is to maximize the margin, that is, the distance between the hyperplane and the closest points of the two classes (a classifier with a wide margin generalizes better)
- *The constraints* require every observation to lie on the right side of the hyperplane — in the soft margin, up to a slack paid in the objective

Data (input of the model) (1/2)

Symbol	Type	Meaning
n	$\in \mathbb{Z}_{\geq 1}$	number of observations of the training set, indexed by $i \in \{1, 2, \ldots, n\}$
p	$\in \mathbb{Z}_{\geq 1}$	number of features, indexed by $j \in \{1, 2, \ldots, p\}$
x_{ij}	$\in \mathbb{Q}$	value of feature j for observation i; $\boldsymbol{x}_i \in \mathbb{Q}^p$ is the vector of observation i

Data (input of the model) (2/2)

Symbol	Type	Meaning
y_i	$\in \{-1, +1\}$	*observed* class label of i (a datum, not a variable)
C	$\in \mathbb{Q}_{>0}$	weight of the margin violations (soft margin only)

- C is upper case to follow the machine learning literature: an exception to the convention that reserves capital letters for sets and matrices

Decision variables

We introduce the following $p + 1$ free variables:

$$\begin{cases} w_j = \text{coefficient } j \text{ of the separating hyperplane} \\ b = \text{intercept of the hyperplane} \end{cases}$$

$$\forall j \in \{1, 2, \ldots, p\}$$

They define the hyperplane

$$\sum_{j=1}^{p} w_j \, x_j + b = 0$$

Hard margin QP model

Hard margin QP model

$$\min \quad \frac{1}{2}\sum_{j=1}^{p} w_j^2 \tag{16.1a}$$

$$\text{subject to} \quad y_i\left(\sum_{j=1}^{p} w_j\, x_{ij} + b\right) \geq 1, \quad \forall i \in \{1, 2, \ldots, n\}, \tag{16.1b}$$

$$w_j \gtreqless 0, \quad \forall j \in \{1, 2, \ldots, p\}, \tag{16.1c}$$

$$b \gtreqless 0. \tag{16.1d}$$

Description of the hard margin QP model

- The convex quadratic objective function minimizes $\frac{1}{2}\|\boldsymbol{w}\|^2$: since the geometric margin between the supporting hyperplanes is $2/\|\boldsymbol{w}\|$, minimizing it is equivalent to *maximizing the margin*
- The linear classification constraints impose that every observation lies on the right side of the hyperplane, at a distance of at least one unit in the scale of $\boldsymbol{w}$ (n linear constraints)
- The last constraints define the variables of the model, all free

Example: Two points, everything by hand (1/3)

- $\boldsymbol{x}_1 = (0, 0)$ with $y_1 = -1$ and $\boldsymbol{x}_2 = (2, 2)$ with $y_2 = +1$

Step 1 — constraints

$$-(b) \geq 1 \qquad \text{and} \qquad 2w_1 + 2w_2 + b \geq 1$$

that is,

$$b \leq -1 \qquad \text{and} \qquad 2(w_1 + w_2) \geq 1 - b \geq 2$$

Example: Two points, everything by hand (2/3)

Step 2 — symmetry and optimum

- To minimize $w_1^2 + w_2^2$ with $w_1 + w_2 \geq 1$ it is best to take $w_1 = w_2 = t$ and the inequalities active:

$$4t + b = 1, \quad b = -1 \quad \Rightarrow \quad t = 1/2$$

- Hence $\boldsymbol{w} = (0.5; 0.5)$ and $b = -1$
- The hyperplane $0.5x_1 + 0.5x_2 = 1$ is the perpendicular bisector of the segment between the two points

Example: Two points, everything by hand (3/3)

Step 3 — margin

$$\frac{2}{\|\boldsymbol{w}\|} = \frac{2}{\sqrt{0.5}} = 2\sqrt{2} = 2.83$$

- It is exactly the distance between the two points
- Both constraints are active: both points are *support vectors*
- With real overlapping data, however, the hard margin is infeasible: the soft margin is needed

Summary

- The hard margin looks for the separating hyperplane of maximum margin
- Margin $= 2/\|\boldsymbol{w}\|$: maximizing it is equivalent to minimizing $\frac{1}{2}\|\boldsymbol{w}\|^2$
- It is a convex QP with n linear constraints and $p+1$ free variables
- The points with an active constraint are the *support vectors*
- With overlapping classes the model is infeasible

Soft margin

Introduction: Soft margin

- With real overlapping data the hard margin is infeasible
- We allow some observations to violate the margin, paying the violation in the objective
- The parameter C governs the trade-off between a wide margin and fit to the data

Slack variables

One adds n non-negative slack variables:

$$\xi_i = \text{margin violation of observation } i, \quad \forall i \in \{1, 2, \ldots, n\}$$

- $\xi_i = 0$: the observation respects the margin
- $\xi_i > 1$: the observation is misclassified

Soft margin QP model (primal)

Soft margin QP model (primal)

$$\min \quad \frac{1}{2}\sum_{j=1}^{p} w_j^2 + C\sum_{i=1}^{n}\xi_i \tag{16.2a}$$

$$\text{subject to} \quad y_i\Big(\sum_{j=1}^{p} w_j\, x_{ij} + b\Big) \geq 1 - \xi_i, \quad \forall i \in \{1,2,\ldots,n\}, \tag{16.2b}$$

$$w_j \gtreqless 0, \qquad \forall j \in \{1,2,\ldots,p\}, \tag{16.2c}$$

$$b \gtreqless 0, \tag{16.2d}$$

$$\xi_i \geq 0, \qquad \forall i \in \{1,2,\ldots,n\}. \tag{16.2e}$$

Description of the soft margin QP model

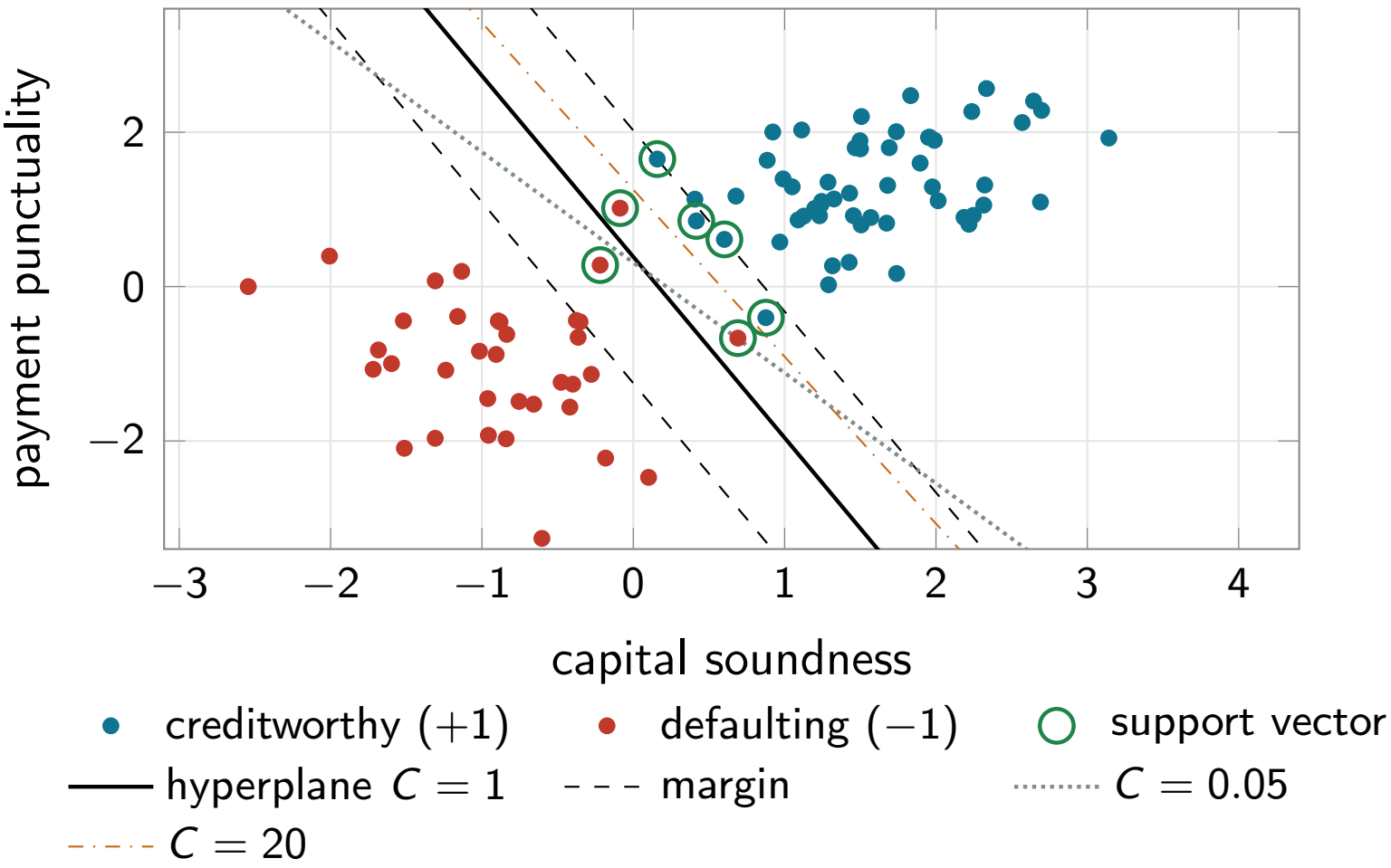
- The convex quadratic objective function balances two terms:
  - the simplicity of the model ($\frac{1}{2}\|\boldsymbol{w}\|^2$, wide margin)
  - the cost of the violations on the training set ($C \sum_i \xi_i$)
- C is the price assigned to the violations
- The linear constraints require correct classification *up to the slack* ξ_i, paid in the objective (n linear constraints)
- Variables: w_j and b free, ξ_i non-negative

Case study: credit risk

- 90 customers, two standardized indicators: capital soundness and punctuality of payments
- 55 creditworthy and 35 defaulting, partially overlapping

C	$\boldsymbol{w}$	b	margin	errors	in the margin
0.05	(0.678; 0.474)	−0.148	2.418	2	22
1.00	(1.433; 0.611)	−0.236	1.284	2	7
20.00	(5.058; 2.338)	−2.936	0.359	0	3

Soft margin: hyperplane, margin, support vectors



How to read the chart

- The 90 customers in the plane of the two indicators: creditworthy (teal), defaulting (red), support vectors circled in green
- The black line is the optimal hyperplane for $C = 1$ with its margin lines (dashed)
- The grey and orange lines are the hyperplanes for $C = 0.05$ (wide margin) and $C = 20$ (narrow margin)

The regularization–fit trade-off

- A small C buys a wide margin while accepting 22 points inside the margin
- A large C chases every point of the training set (zero errors, a very narrow margin)
- This is the **regularization–fit** trade-off: C must be chosen with validation data, never by looking at the test set
- The three lines are similar but not equal: the slope changes because the "paid for" points change

Summary

- The soft margin adds the slacks $\xi_i \geq 0$, each paid C in the objective
- It remains a convex QP, always feasible
- Small C: wide margin, more violations; large C: narrow margin, fit to the training set
- C is chosen with validation data, never with the test set

The dual formulation and the support vectors

Introduction: The dual formulation

- We associate a multiplier with every classification constraint and write the dual of the soft margin
- The dual reveals which observations really determine the boundary: the **support vectors**
- The dual depends on the data only through inner products: it is the door to kernels

Multipliers

By associating a multiplier α_i with each classification constraint of the soft margin, one obtains n new variables:

$$\alpha_i = \text{multiplier of the classification constraint of observation } i$$

$$\forall i \in \{1, 2, \ldots, n\}$$

Dual QP model of the soft margin

Dual QP model of the soft margin

$$\max \quad \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \, \alpha_j \, y_i \, y_j \, \boldsymbol{x}_i' \boldsymbol{x}_j \tag{16.3a}$$

$$\text{subject to} \quad \sum_{i=1}^{n} \alpha_i \, y_i = 0, \tag{16.3b}$$

$$\alpha_i \leq C, \quad \forall i \in \{1, 2, \ldots, n\}, \tag{16.3c}$$

$$\alpha_i \geq 0, \quad \forall i \in \{1, 2, \ldots, n\}. \tag{16.3d}$$

Description of the dual model

- The concave quadratic objective function depends on the data *only* through the inner products $\boldsymbol{x}_i'\boldsymbol{x}_j$: this is the observation that opens the door to the kernel
- The linear constraint $\sum_i \alpha_i y_i = 0$ follows from stationarity with respect to the intercept b (one linear constraint)
- The constraints $\alpha_i \leq C$ bound each multiplier by the price C of the violations (n linear constraints)
- The constraints $\alpha_i \geq 0$ define the variables of the model

Primal-dual relation

$$\boldsymbol{w} = \sum_{i=1}^{n} \alpha_i \, y_i \, \boldsymbol{x}_i \qquad\qquad f(\boldsymbol{x}) = \operatorname{sign}(\boldsymbol{w}'\boldsymbol{x} + b)$$

On the credit risk case ($C = 1$):

- $\boldsymbol{w}$ from the dual $= (1.433;\ 0.611)$, $b = -0.236$: identical to the primal
- dual value 4.7868 = primal value (strong duality)

multiplier	points	meaning
$\alpha_i = 0$	83	do NOT affect the boundary
$0 < \alpha_i < C$	2	exactly on the margin
$\alpha_i = C$	5	inside the margin or misclassified

Why they are called support vectors (1/2)

- The solution depends *only* on the 7 points with $\alpha_i > 0$: it is they that "support" the hyperplane
- The other 83 could disappear without changing anything
- Moreover, the dual depends on the data only through the inner products $\boldsymbol{x}_i'\boldsymbol{x}_j$: this observation opens the door to the kernel

Why they are called support vectors (2/2)

The KKT complementarity organizes everything:

- $\alpha_i = 0 \Rightarrow$ point beyond the margin
- $0 < \alpha_i < C \Rightarrow$ exactly on the margin (from these one recovers b)
- $\alpha_i = C \Rightarrow$ the constraint "pays" a slack $\xi_i > 0$

Summary

- The dual has one variable $\alpha_i \in [0, C]$ per observation
- $\boldsymbol{w} = \sum_i \alpha_i y_i \boldsymbol{x}_i$: the boundary depends only on the points with $\alpha_i > 0$, the support vectors
- Strong duality: dual value = primal value
- The data enter the dual only through the inner products $\boldsymbol{x}_i' \boldsymbol{x}_j$

Kernels: non-linear boundaries from a convex QP

Introduction: Kernels

- Not all classes can be separated by a straight line
- Exploiting the dual, we replace the inner product with a **kernel**
- We obtain curved boundaries while staying in a convex QP

The RBF kernel

- We replace $\boldsymbol{x}_i'\boldsymbol{x}_j$ with a kernel $K(\boldsymbol{x}_i, \boldsymbol{x}_j)$, for example the Gaussian RBF (*Radial Basis Function*) kernel

$$K(\boldsymbol{x}, \boldsymbol{z}) = e^{-\gamma\|\boldsymbol{x}-\boldsymbol{z}\|^2}$$

- One classifies in a transformed space without ever building it:

$$f(\boldsymbol{x}) = \operatorname{sign}\left(\sum_{i=1}^{n} \alpha_i y_i K(\boldsymbol{x}_i, \boldsymbol{x}) + b\right)$$

- The boundary can be arbitrarily curved, but training *remains a convex QP* if the kernel is valid (positive semidefinite matrix)

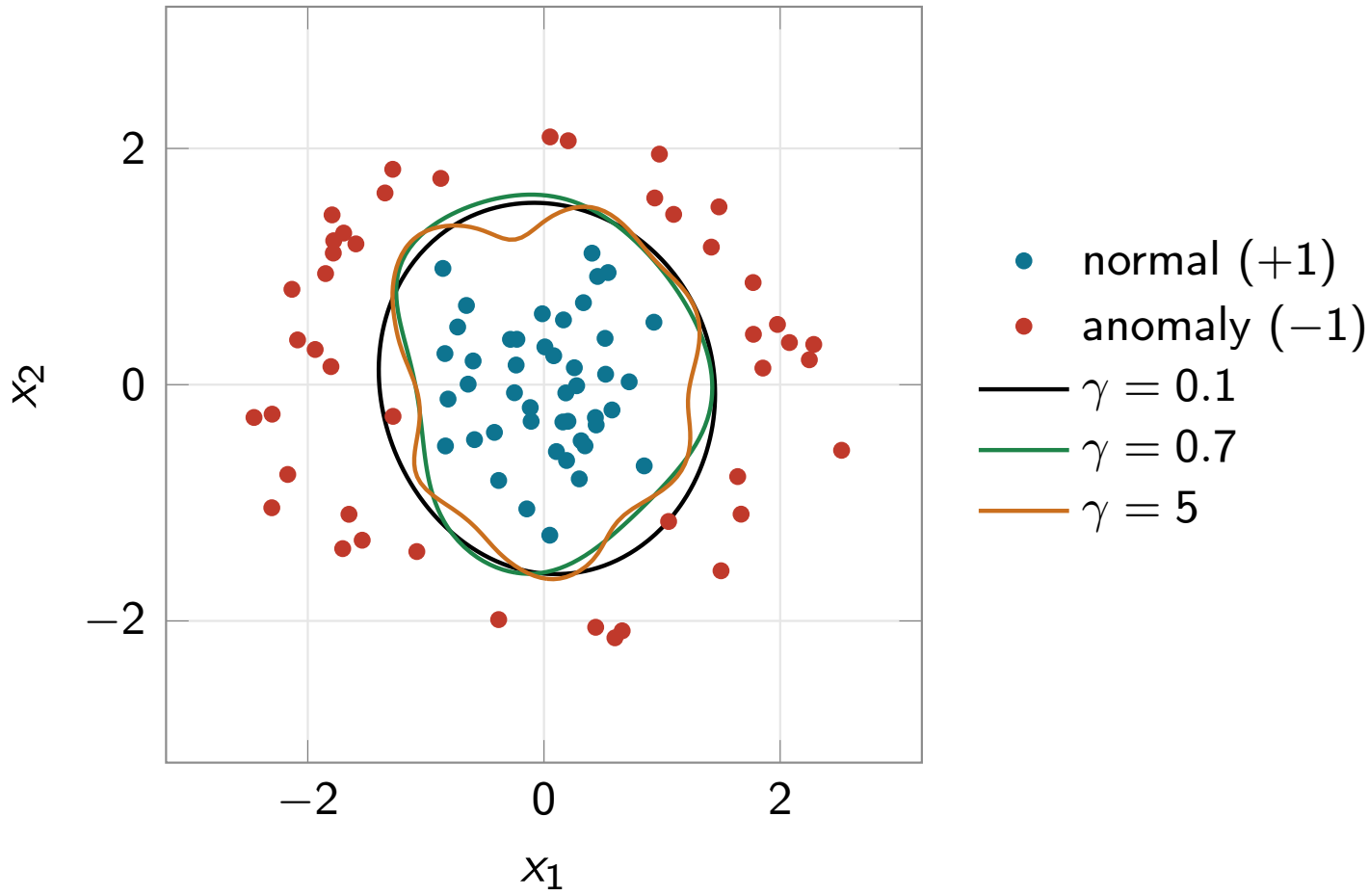
Case study: "ring-shaped" anomalies

- 90 process observations
- Normal operation in the centre, the anomalies arranged in a ring
- No straight line can separate them

γ	training errors	support vectors
0.1	1	20
0.7	0	15
5.0	0	61

With $\gamma = 5$ the model "memorizes" the points.

RBF kernel: the level-0 boundary



How to read the chart

- The "ring-shaped" dataset: normal operation in the centre, anomalies in a ring
- The curves are the decision boundaries of the RBF kernel for three values of γ
- With $\gamma = 0.1$ the boundary is a regular circle
- With $\gamma = 5$ it breaks up around the individual points (overfitting)

Overfitting

Observation 1: Convexity does not mean generalization

With $\gamma = 5$ the boundary breaks up into islands around the individual points and the support vectors rise to 61 out of 90: the model is memorizing the training set (overfitting), while still remaining a convex QP.

Convexity of training and the ability to generalize are different properties. C and γ must be chosen together by cross-validation.

Summary

- The kernel replaces the inner product in the dual: non-linear boundaries without building the transformed space
- With a valid kernel training remains a convex QP
- Large γ: fragmented boundary, many support vectors, overfitting
- C and γ are chosen together by cross-validation

Unbalanced classes and asymmetric costs

Introduction: Unbalanced classes

- In managerial applications the important class is rare
- An error on the rare cases costs more (an undetected defaulter, an undiscovered fraud)
- We see how to build asymmetric costs into the model

Weighting the slacks

It is enough to weight the slacks differently for the two classes:

$$C_- \sum_{i:\, y_i=-1} \xi_i \; + \; C_+ \sum_{i:\, y_i=+1} \xi_i$$

	precision	recall defaulters
equal costs ($C = 1$)	1.00	0.94
$10\times$ cost on the defaulters	0.97	1.00

Decision costs, not frequencies

- With the 10× cost the boundary moves back towards the "healthy" class
- No defaulter escapes (recall 100%) at the price of a few more false alarms (precision from 100% to 97%)
- The weights must reflect the *decision costs*: how much a loan granted to someone who will not repay costs vs. a good customer turned down
- Not the frequencies of the classes

Summary

- With unbalanced classes one uses two prices C_- and C_+ for the slacks
- The model remains a convex QP
- One evaluates with precision and recall, not with accuracy alone
- The weights reflect the decision costs, not the frequencies

Support Vector Regression

Introduction: Support Vector Regression

- The same idea applies to estimating a continuous function
- SVR ignores the errors within a tube of width ε
- Only the points outside the tube determine the estimate

The SVR model

$$\min \ \tfrac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*)$$

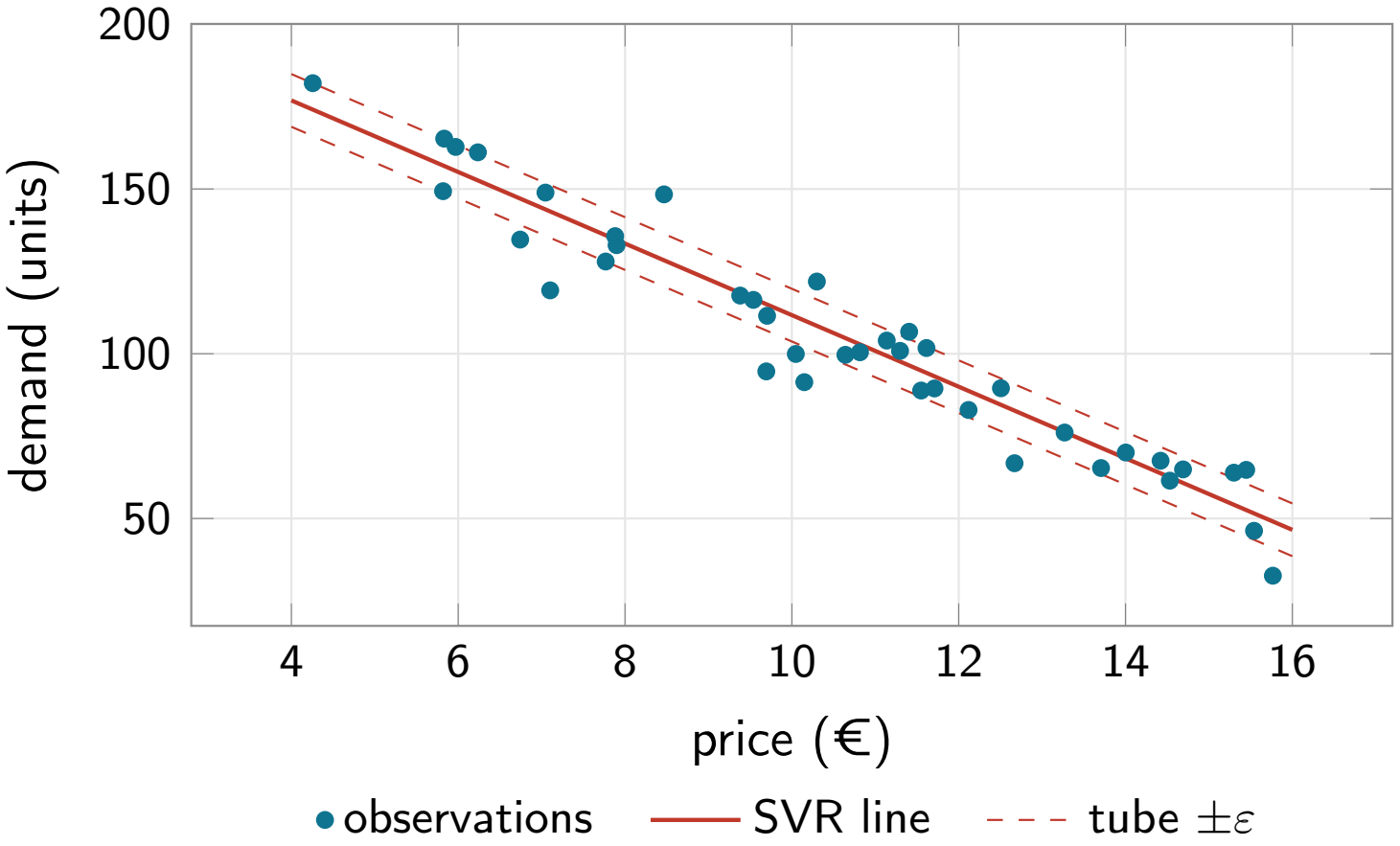
with

$$|y_i - (\boldsymbol{w}'\boldsymbol{x}_i + b)| \le \varepsilon + \xi_i^{(*)} \qquad \text{for every } i$$

On the price–demand relation (40 observations):

- SVR line: demand $= -10.86 \cdot \text{price} + 220.32$ (true: $-11p + 220$)
- tube $\varepsilon = 8$: 14 points out of 40 outside the tube $\rightarrow$ only they determine the line

SVR: only the points outside the tube pay a penalty



How to read the chart

- The 40 observations of the price–demand relation
- The estimated line and the tolerance tube $\pm\varepsilon = 8$
- Only the 14 points outside the tube pay a penalty and determine the line

Summary

- SVR estimates a continuous function with a convex QP
- Errors within the tube $\pm\varepsilon$ are not paid
- Only the points outside the tube determine the line: they are the support vectors of regression

Exercises

Introduction: Exercises

- The exercises put the concepts of the chapter to the test
- They are solved by hand or with the lab script and Gurobi
- They cover hard margin, complementarity, the choice of C, the decision threshold and the One-Class SVM

Exercise: check

- Solve by hand the hard margin with

$$\boldsymbol{x}_1 = (0, 0),\ y_1 = -1 \qquad \text{and} \qquad \boldsymbol{x}_2 = (4, 0),\ y_2 = +1$$

- Check with Gurobi:

$$\boldsymbol{w} = (0.5;\ 0), \qquad b = -1, \qquad \text{margin } 4$$

Exercise: complementarity

- For $C = 1$, list from the script the 7 points with $\alpha_i > 0$
- Check numerically the three complementarity conditions:

$$\alpha_i = 0 \Rightarrow \xi_i = 0 \text{ and margin } \geq 1; \quad \text{and so on}$$

Exercise: validation curve

- Split the 90 customers into training (60) and validation (30)
- Plot the validation error for $C \in \{0.01; 0.1; 1; 10; 100\}$ and choose C
- Is the training error monotone in C? And the validation error?

Exercise: decision threshold

- For the same model ($C = 1$), move the threshold: classify as defaulting if

$$\boldsymbol{w}'\boldsymbol{x} + b < \theta, \qquad \theta \in [-1, 1]$$

- Plot precision and recall as a function of θ
- Where is the point that maximizes profit if an undetected defaulter costs 10 and a false alarm 1?

Exercise: one-class

- Implement the One-Class SVM on the ring-shaped dataset using only the normal points:

$$\min \ \tfrac{1}{2}\|\boldsymbol{w}\|^2 + \tfrac{1}{\nu n}\sum_{i=1}^{n}\xi_i - \rho \quad \text{subject to} \quad \boldsymbol{w}'\phi(\boldsymbol{x}_i) \geq \rho - \xi_i$$

- RBF kernel, $\gamma = 0.7$
- Which fraction of anomalies does it detect for $\nu \in \{0.05;\ 0.1;\ 0.2\}$?

Final summary (1/2)

Models:

- Hard margin: convex QP, maximizes the margin $2/\|\boldsymbol{w}\|$
- Soft margin: slacks ξ_i paid C, always feasible
- Dual: $\alpha_i \in [0, C]$, $\boldsymbol{w} = \sum_i \alpha_i y_i \boldsymbol{x}_i$

Support vectors:

- Only the points with $\alpha_i > 0$ determine the boundary
- KKT complementarity says where each point lies

Final summary (2/2)

Extensions:

- Kernels: non-linear boundaries from a convex QP
- Asymmetric costs C_-, C_+ for unbalanced classes
- SVR: regression with a tolerance tube $\pm\varepsilon$

Modeller's choices:

- C and γ with validation data, never with the test set
- The weights reflect the decision costs, not the frequencies